[INCH-POUND]
A-A-59478
30 September 1999
SUPERSEDING
MIL-P-288B
12 May 1982

COMMERCIAL ITEM DESCRIPTION

PROTRACTOR, THREE ARM

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-P-288B for all federal agencies.

1. SCOPE. This commercial item description (CID) covers two types and three classes of protractors with three movable arms used in position, plotting, and hydrographic surveying.

2. CLASSIFICATION. The types and classes of protractors provided shall be as follows (see 7.2(b)).

Type I - Metal
Class 1 - Three arm, left hand, circular
Class 2 - Three arm, right hand, circular

Type II - Plastic
Class 1 - Three arm, bilateral, circular

3. SALIENT CHARACTERISTICS

3.1 Type I protractors. All materials and hardware used in the construction of type I protractors shall be corrosion resistant. The assembled protractor shall consist of a circle frame with an inlaid graduated circle and three arms, one of which shall be fixed and attached to the circle frame and two that shall be movable and attached to a pivot center. A tangent screw with vernier scale shall be attached to each of the two movable arms. A reading glass shall be hinged to a rotatable collar, attached to the pivot center. Figure 1 is an example of an acceptable design of the ribs for the circle frame.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 6675

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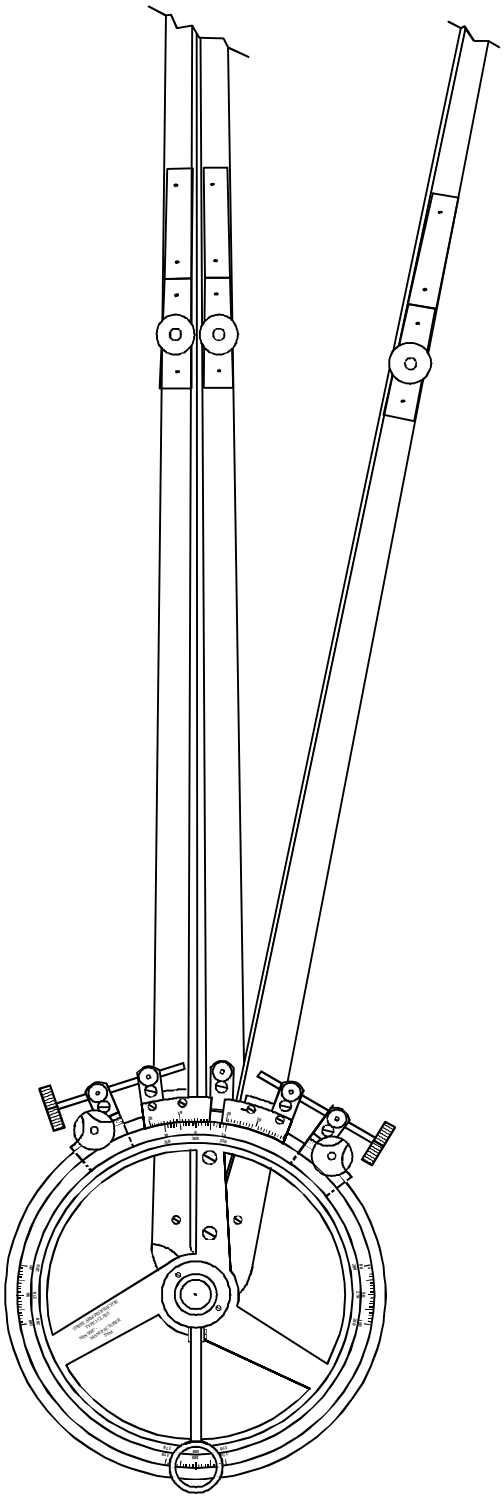


FIGURE 1. Type I protractor.

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3.1.1 Accuracy. The graduations on the body of the type I protractors shall be such that no graduation mark will vary from its true position by more than 2 minutes of arc. The movable arms shall be mounted in the assembled instrument so as to rotate on an axis at the center of the graduated circle, with an accuracy such that the maximum error due to eccentricity shall not exceed 3 minutes of arc. The total error of the assembled protractor shall not exceed 5 minutes of arc.

3.1.2 Circle frame. The circle frame shall have a numbered scale engraved on the top surface of the frame such that, when assembled with the graduated circle, numerals will appear at every 10 degree graduation starting at 360 degrees at the fiducial point and continue clockwise in decreasing increments of 10 to 180 degrees, then in increasing increments of 10 to 350 degrees. Numerals shall be 0.070 inch high and shall be filled with black enamel. The base of the numerals shall be toward the center of the circle. The frame shall have a circular groove on the undersurface to receive the guide pins of the clamps. The center of the circle frame shall be a disc not greater than 1-7/16 inches in outside diameter. It shall be connected to the outer circle of the frame by three radial ribs. The center, ribs, and outer circle of the frame shall be fabricated as one integral piece and shall form a rigid support for the arms. The center of the circle frame shall be fitted for the pivot center. The required identification information shall be located on the top rib of the circle frame (see 6.2).

3.1.3 Graduated circle. The graduated circle shall be not less than one-half inch wide, with the outside diameter not less than 6-5/16 inches and the inside diameter not greater than 5-7/32 inches. The entire circle shall be accurately graduated at the outer circumference, in degrees and half degrees. The graduated lines shall have a depth of 0.003 inch minimum to 0.005 inch maximum, shall have a width of 0.005 ± 0.001 inch and shall be filled with waterproof ink or enamel. The length of the graduated lines shall be as follows: 10 degree marks 0.19 inch, 5 degree marks 0.15 inch, 1 degree marks 0.11 inch, and 30 minute marks 0.07 inch. Numerals shall be engraved every 10 degrees in both a clockwise and counter-clockwise direction, starting with zero at the fiducial point and continuing to 180 degrees. The base of the numerals shall be toward the center of the circle.

3.1.4 Pivot center. The pivot center shall be secured to the center of the circle frame by countersunk screws or other suitable means. The pivot center shall have a projection above the center disc of the frame for retaining the collar of the reading glass, and a projection beneath the center disc for retaining the collars of the two movable arms. The inside and outside surfaces of the pivot center shall be true cylinders, concentric with the circle of the instrument, and shall be ground to an accurate fit for the collars. The pivot shall be free from imperfections and finished smooth.

3.1.5 Pivot plugs. Four pivot plugs shall be furnished for use in marking positions. They shall be free from imperfections and finished smooth. The outside diameter of the plugs shall allow for a sliding fit in the pivot center. The four kinds of plugs shall be as follows:

a. A cylindrical hollow plug, with a glass or clear plastic bottom, having a hole in the center to receive a pencil point and marked with cross lines at right angles on the bottom plastic

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surface. The glass or plastic bottom shall be set flush in the hollow plug. The hollow plug shall be painted with white enamel on the inside.

b. A solid cylindrical plug with a plunger having a point on its lower end, fitted with a spring to hold the point clear of the bottom when not pressed from above.

c. A cylindrical plug, with a glass or clear plastic bottom having cross lines on the undersurface, at right angles and intersecting accurately at the center on the bottom surface. The glass or plastic bottom shall be set flush in the hollow plug to avoid parallax. The hollow plug shall be painted with white enamel on the inside.

d. A solid cylindrical plug having a short needle point in the center of the base.

3.1.6 Arms. The arms shall be flat bars having a length of 18 inches minimum to 18-3/4 inches maximum from the center point of the instrument. Each arm shall be one-eighth inch in thickness with a top-edge bevel angle of 20 to 45 degrees along the radial straightedge and a face width of one thirty-second to one-sixteenth of an inch. Each arm shall be 0.6 inch in width at its outer extremity and shall be tapered to accomplish the following construction. The fixed arm shall be secured, by means of screws and pins, to a rib of the circle frame in such a position that the beveled edge will lie on the radius of the circle, which passes through the fiducial point on the graduated circle. The two movable arms shall be secured to collars which rotate about the pivot center. Each movable arm shall be fitted with a vernier, clamp, and tangent screw. The collars shall be reduced in thickness, so that when the arms are secured in place they will come flush with the bottom of the pivot center and lie on the same plane.

3.1.7 Straightedges, type I, class 1. Using the center point as datum, the straightedge of the left movable arm for the type I, class 1 protractor shall lie on a radius of the circle and shall pass through the fiducial point of the graduated circle when the left arm of the protractor is closed. The straightedge of the right movable arm for the type I, class 1 protractor shall lie on a radius of the circle and shall be at an angle not greater than 11 degrees, 30 minutes from the fiducial point, when the right arm of the protractor is closed.

3.1.8 Straightedges, type I, class 2. Using the center point as datum, the straightedge of the right movable arm for the type I, class 2 protractor shall lie on a radius of the circle and shall pass through the fiducial point of the graduated circle when the right arm of the protractor is closed. The straightedge of the left movable arm for the type I, class 2 protractor shall lie on a radius of the circle and shall be at an angle not greater than 11 degrees, 30 minutes from the fiducial point when the left arm of the protractor is closed.

3.1.9 Angle range, 260 degrees. The straightedge of the left movable arm for the type I, class 1 protractor and the straightedge of the right movable arm for the type I, class 2 protractor shall be capable of laying off angles up to 260 degrees.

3.1.10 Angle range, 270 degrees. The straightedge of the right movable arm for the type I, class 1 protractor and the straightedge of the left movable arm for the type I, class 2 protractor shall be capable of laying off angles up to 270 degrees.

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3.1.11 Extension arms. Three extension arms shall be provided. Each extension arm shall be 13-1/2 inches long, tapering in width from 0.6 to 0.4 inch at the outer end, and shall be joined to the permanent arms by scarfing pieces. The scarfing pieces shall be permanently attached to the extension arms and secured, when needed, to the permanent arms by means of a threaded thumbscrew and two dowel pins affixed to the scarfing pieces. Extension arms shall be match-marked to the mating arm.

3.1.12 Vernier scales. The upper surface of the vernier scales shall be flush with the surface of the graduated circle. The vernier of each movable arm shall have its zero graduation positioned over the straightedge of the arm. Each vernier shall embrace 14 degrees, 30 minutes of the graduated circle and shall be so graduated as to permit reading of the scale to l minute of arc. Each vernier shall have one additional mark on each end of the scale. The graduation requirements shall be the same as those of the graduated circle (see 3.1.3). The engraved numerals shall be numbered 0, 10, 20, and 30.

3.1.13 Clamps. Each clamp shall be secured to the circle frame by a thumbscrew and guided in motion by a tongue which slides in the circular groove in the underside of the circle frame.

3.1.14 Tangent screws. The tangent screws shall be of an appropriate size and have knurled or otherwise slip-resistant heads. Each tangent screw shall be attached to its clamp by a fixed ball working in a socket joint, and to the movable arm by a ball nut working in a socket joint.

3.1.15 Reading glass. The reading glass shall be a 3.0X single-lens magnifier having a focal length of 1-1/2 inches and a clear aperture of five-eighths inch. It shall be hinged to a collar which rotates around the pivot center so as to have motion parallel to the plane of the instrument. The field of view of the reading glass, when the glass is parallel to the plane of the instrument, shall be centered over the graduated edge of the graduated circle and the vernier scale.

3.1.16 Magnifying glass. A pocket magnifying glass of 3.0X magnifying power shall be furnished with each type I protractor.

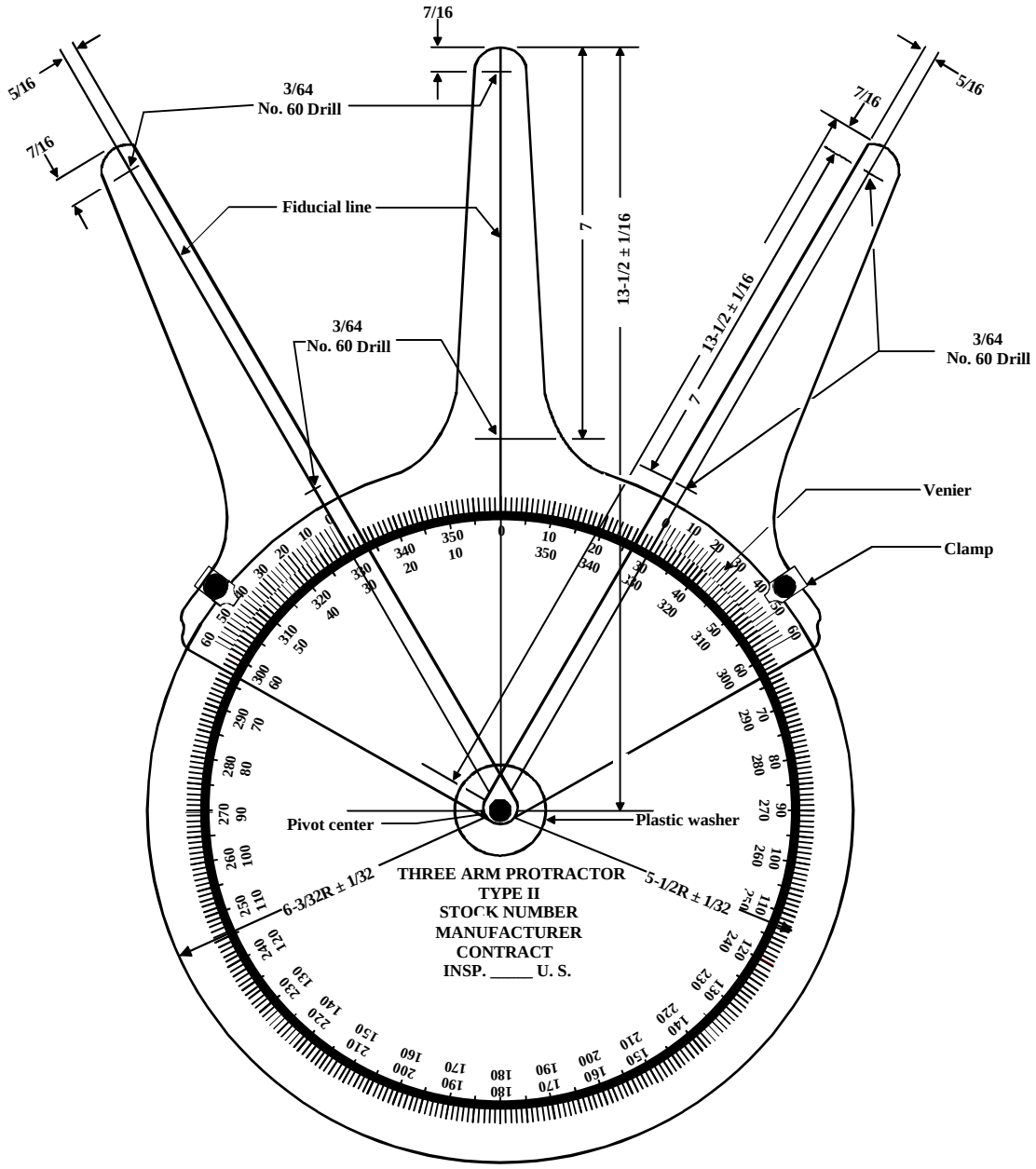
3.1.17 Screwdriver. A screwdriver of good commercial quality, of a size fitting the screws on the instrument, shall be furnished with each type I protractor.

3.2 Construction, type II protractor.

3.2.1 Assembly. All materials and hardware used in the construction of type II protractors shall be corrosion resistant. The assembled protractor shall consist of a body having a graduated circle with a fixed integral arm, a pivot center, two movable arms with vernier scales, and thumb nut clamps on each movable arm (see figure 2).

3.2.2 Accuracy. The total error of the assembled protractor shall not exceed 6 minutes of arc.

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All dimensions are in inches

FIGURE 2. Type II protractor.

3.2.3 Body. The body shall be made of a high-polish finished plastic 0.040 inch (nominal) thick. The body shall be a circle 12-3/16 inches in diameter (plus or minus one-sixteenth inch) with a fixed integral arm extending 13-1/2 inches (plus or minus one-sixteenth inch) from the center of the circle. The arm shall taper from a width of 2-1/2 inches measured on the circumference of the circle to a width of approximately 1-inch at the rounded end. A fiducial

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line shall extend from the pivot center through the center of the fixed arm for the full length of the arm. Two holes, three sixty-fourths inch in diameter, shall be drilled through the fiducial line, one at seven-sixteenths of an inch from the outer end, and the other at 7 inches from the outer end of the arm. The circle shall be accurately graduated on the upper surface (see 3.2.5). The required identification information shall be located on the underside of the body (see 6.2).

3.2.4 Movable arms. There shall be two movable arms made of a high-polish finished plastic 0.030 inch (nominal) thick, which shall pivot about the center of the body. Both movable arms shall be the same length as the fixed arm and shall be rounded on the outer ends. Each movable arm shall have a fiducial line extending radially from the pivot center through the greatest length of the arm. Each arm shall have a smooth straightedge parallel to and five-sixteenths of an inch distance from the fiducial line. The smooth parallel edge of one arm shall be to the left of the fiducial line and on the other arm, to the right of the fiducial line, when the two fiducial lines are in coincidence. The shape of each arm on the opposite side to the smooth parallel edge shall be such that the arm covers approximately 33 degrees of the protractor graduations, when the fiducial lines of the movable arms are in coincidence with the fiducial line of the fixed arm. This side of the arms shall extend past the periphery of the protractor body a distance of three-eighths of an inch, to accommodate a clamp (see 3.2.7), and shall taper toward the outer end at such an angle that the total width of the arm at the outer end is approximately 1 inch. Two holes, three sixty-fourths inch in diameter, shall be drilled through the fiducial line, one seven-sixteenths inch from the outer end and the other 7 inches from the outer end of each arm. When the fiducial lines of the two movable arms are in coincidence with the fiducial line of the fixed arm, their holes shall coincide with the holes in the fixed arm. The movable arms shall be accurately graduated on the underside to serve as verniers (see 3.2.5).

3.2.5 Graduations. The protractor body shall be accurately graduated and numbered on the upper surface. The graduations shall be either engraved or printed with a tolerance of plus or minus 5 minutes of arc. The outside diameter of the graduations shall be 11 inches (plus or minus one-sixteenth inch). The fiducial line of the fixed arm shall form the zero line, and the graduations shall be marked in 1-degree increments and numbered every 10 degrees. Numerals shall have their base toward the center and shall read from 0 to 350 degrees in two rows of numerals, one in a clockwise and one in a counterclockwise direction. Clockwise numerals shall form the outside row. The movable arms shall be accurately graduated and numbered on the underside, such that the numerals can be read correctly from the top side. The graduations on the movable arms shall serve as verniers, readable to 2 minutes of arc. They shall be numbered every fifth graduation, starting with zero at the fiducial line and continuing from 10 through 60. The base of the vernier numerals shall be toward the center of the body. The vernier graduations shall be concentric with the circle graduations and shall have their base in coincidence with the outside diameter of the graduated circle. All graduation marks and fiducial lines shall be 0.004 to 0.010 inch wide. The length of the graduation marks shall be as follows: 1 degree and 2 minute marks five thirty-seconds of an inch; 5 degree and 10 minute marks one-quarter inch; 10 degree marks five-sixteenths of an inch. Numerals on the body shall be five thirty-seconds of an inch high and on the movable arms, one-eighth inch high. All graduation marks, fiducial lines, and numerals shall be filled or printed with permanent waterproof black ink or black enamel.

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They shall be protected with a transparent plastic cover sheet from 0.003 to 0.005 inch thick of a material compatible with the base material.

3.2.6 Pivot center. The pivot center shall consist of a center post, knurled or otherwise slip-resistant ring nut, and plastic washer arranged to allow for free axial movement of the movable arms. The center post and ring nut shall have an overall height not to exceed one-quarter inch when assembled. The bottom of the post shall have a flange three-eighths inch in diameter which shall project one thirty-second of an inch below the underside of the protractor body. The edges of the flange shall be slightly rounded in order to facilitate sliding of the protractor over the chart surface. The stem of the post shall be one-quarter inch in diameter and shall be threaded to receive the ring nut. The top of the post shall have an opening approximately three-sixteenths of an inch in diameter and counterbored to within one thirty-second of an inch of the bottom. A three sixty-fourths inch diameter hole shall be drilled through the bottom center of the post. A plastic washer (see 3.4.2) of 1-1/2 inch diameter shall be provided over the two movable arms. A ring nut shall be provided to secure the washer and movable arms, yet allow for free axial movement of the arms.

3.2.7 Clamps. Each movable arm shall be provided with a U-shaped clamp made of brass or other suitable material. The bottom of each clamp shall have a threaded post secured so that it will not turn. The top of the clamps shall be shaped at the inner edge so that they will not obstruct the view of any of the numerals or graduations on the vernier. A black knob, with a threaded insert made of brass or other suitable material, shall be used for tightening the clamp. A washer, made of brass or other suitable material, shall be provided between the knob and the clamp. This washer shall be approximately one thirty-second of an inch thick.

3.3 Storage and carrying case.

3.3.1 Type I protractor. The storage and carrying case for type I protractors shall have a permanently attached cover to protect the instruments and accessories, and shall provide a means of securing them inside. The case, fittings, fasteners, and information plate shall be corrosion resistant, and the case shall be lined with a fungus-resistant, non-abrasive cushioning material. The case shall be only large enough to contain the closed instrument, three extension arms, three thumbscrews for extension arms, one magnifying glass, one screwdriver and four pivot plugs. Each case shall be identified with an information plate attached to the lower front face of the case (see 6.1).

3.3.2 Type II protractor. Each type II protractor shall be provided with a folding type cardboard or paperboard envelope or folder. The envelope or folder shall be marked with the information required (see 6.1).

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product as that which the supplier offers for sale on the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The company producing the protractor shall have been producing this product which meets the requirements of this CID for at least two years.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(c)). In the absence of detailed packaging requirements, the American Society for Testing and Materials (ASTM) standard D 3951, "Standard Practice for Commercial Packaging", shall be used.

6.2 Special markings. Each protractor shall be identified legibly and permanently (see 7.2(d)). Location of identification shall be: Type I on the top of a rib of the circle frame; type II on the underside of the body such that it can be correctly read from the top, below the pivot point.

7. NOTES

7.1 Sources of documents

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.1.2 ASTM standards. Copies of ASTM Standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2 Ordering data. Acquisition documents must specify the following information:

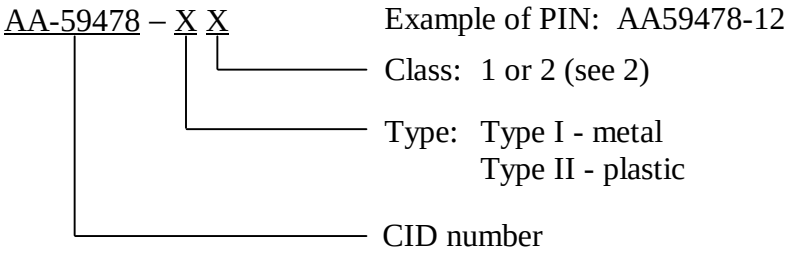
a. Title, number, and date of this document
b. Type and class of protractor required (see 2)
c. Packaging requirements (see 6.1)
d. Special markings (see 6.2)

7.3 National stock numbers (NSNs). The NSNs listed below are assigned for the listed product types and sizes (see 2). Other NSNs may also correspond with this document.

Assigned NSN	Type	Class
6675-00-191-1514	II	1
6675-00-191-1515	I	1
6675-00-191-1516	I	2

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7.4 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.



7.5 Sources of supply. The manufacturers and/or suppliers listed below are known to supply products which meet the salient characteristics requirements of this document. Competition is not limited to the listed firms.

Theodore Alteneder & Son
439 Main Street
Darby, PA 19023-2629
(610) 522-9444

Warren-Knight Instrument Co.
2045 Bennett Road
Philadelphia, PA 19116-3019
(215) 464-9300

MILITARY INTERESTS:

Custodians:
Air Force -99
Army - CE3
Navy - SH

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FCGC

Preparing activity:
DLA - GS
(Project 6675-0085)

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